



# IN241 Statistika

## *Lecture 02*

Dosen:

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## Lecture 02

# Overview and Descriptive Statistics

**\*Based on textbooks of [Ross , 2004] and [Devore, 2016].**

# Populations, Samples, and Processes

# Populations and Samples

A *population* is a well-defined collection of objects.

When information is available for the entire population we have a *census*. A subset of the population is a *sample*.

# Data and Observations

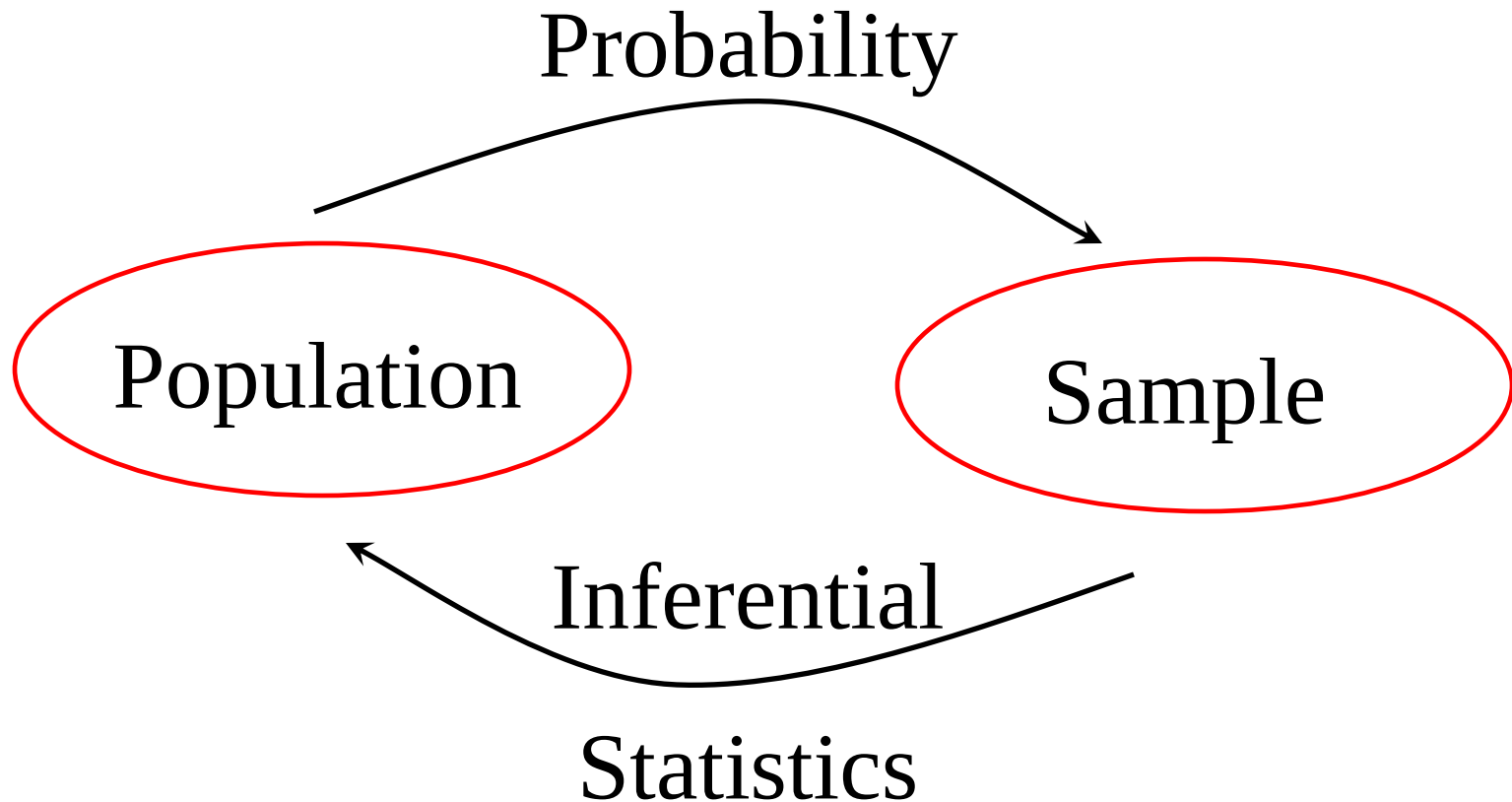
*Univariate* data consists of observations on a single variable (*multivariate* – more than two variables).

# Branches of Statistics

*Descriptive Statistics* – summary and description of collected data.

*Inferential Statistics* – generalizing from a sample to a population.

# Relationship Between Probability and Inferential Statistics



# Pictorial and Tabular Methods in Descriptive Statistics



# Stem-and-Leaf Displays

1. Select one or more leading digits for the stem values. The trailing digits become the leaves.
2. List stem values in a vertical column.
3. Record the leaf for every observation.
4. Indicate the units for the stem and leaf on the display.

# Stem-and-Leaf Example

Observed values:

9, 10, 15, 22, 9, 15, 16, 24, 11

|   |  |           |
|---|--|-----------|
| 0 |  | 9 9       |
| 1 |  | 1 0 5 5 6 |
| 2 |  | 2 4       |

Stem: tens digit

Leaf: units digit

# Stem-and- Leaf Displays

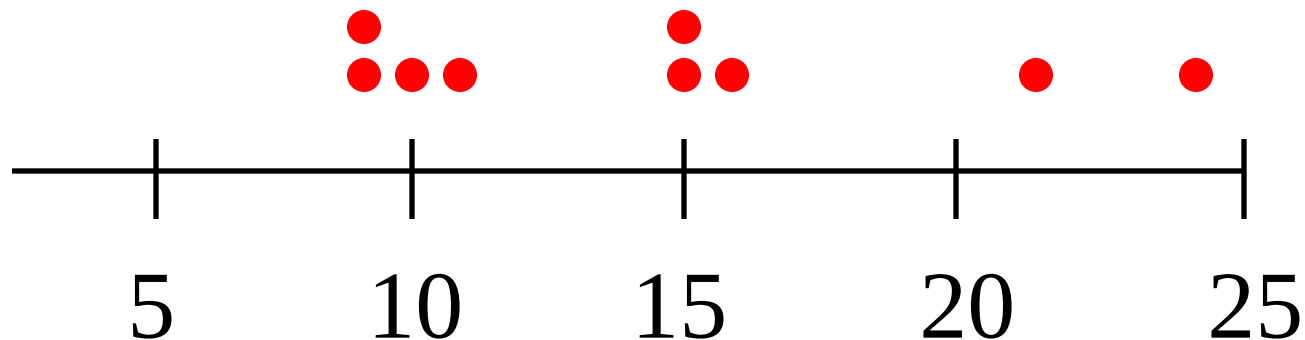
- Identify typical value
- Extent of spread about a value
- Presence of gaps
- Extent of symmetry
- Number and location of peaks
- Presence of outlying values

# Dotplots

Represent data with dots.

Observed values:

9, 10, 15, 22, 9, 15, 16, 24, 11



# Types of Variables

A variable is *discrete* if its set of possible values constitute a finite set or an infinite sequence. A variable is *continuous* if its set of possible values consists of an entire interval on a number line.

# Histograms

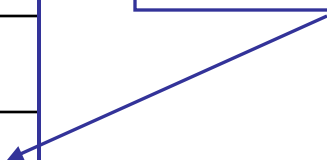
## Discrete Data

Determine the frequency and relative frequency for each value of  $x$ . Then mark possible  $x$  values on a horizontal scale. Above each value, draw a rectangle whose height is the relative frequency of that value.

**Ex.** Students from a small college were asked how many credit cards that they carry.  $x$  is the variable representing the number of cards and the results are below.

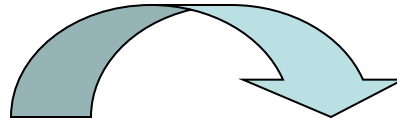
| $x$ | #people | <i>Rel. Freq</i> |
|-----|---------|------------------|
| 0   | 12      | 0.08             |
| 1   | 42      | 0.28             |
| 2   | 57      | 0.38             |
| 3   | 24      | 0.16             |
| 4   | 9       | 0.06             |
| 5   | 4       | 0.03             |
| 6   | 2       | 0.01             |

Frequency  
Distribution

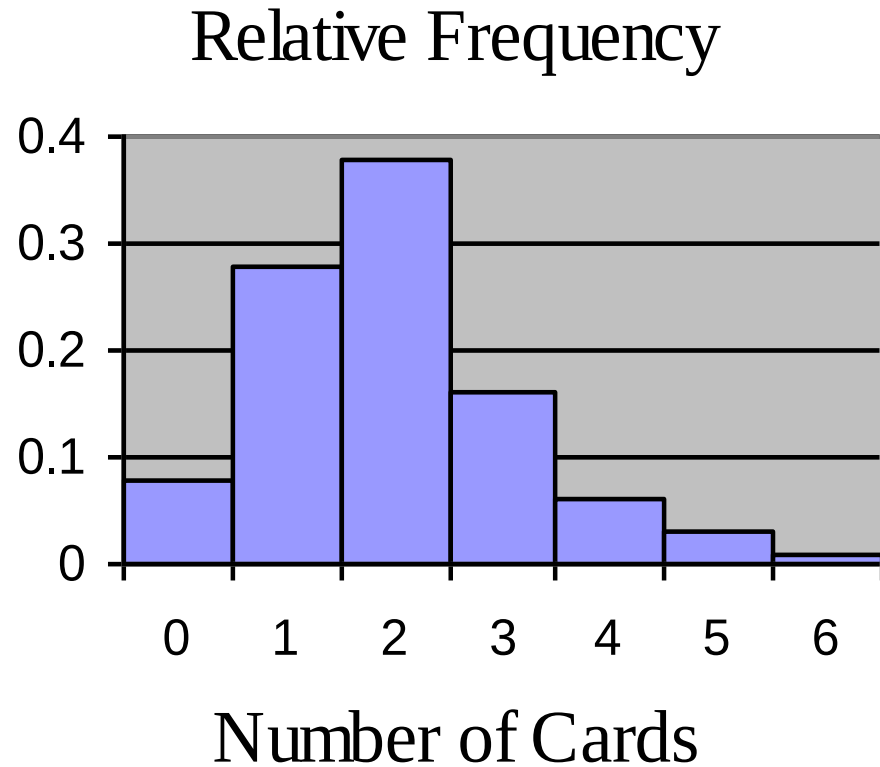


# Histograms

Credit card results:

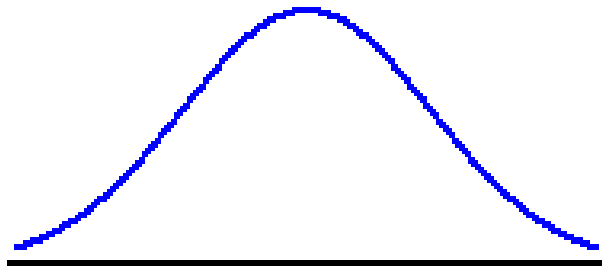


| $x$ | <i>Rel. Freq.</i> |
|-----|-------------------|
| 0   | 0.08              |
| 1   | 0.28              |
| 2   | 0.38              |
| 3   | 0.16              |
| 4   | 0.06              |
| 5   | 0.03              |
| 6   | 0.01              |

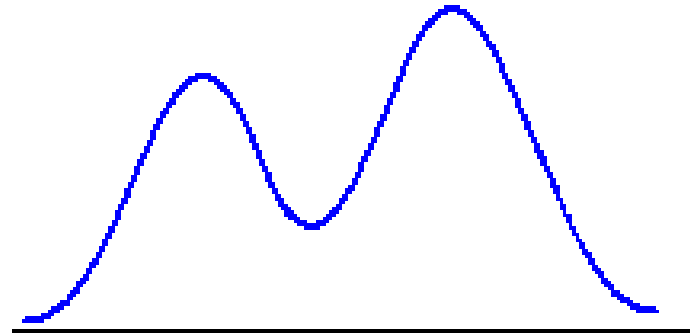




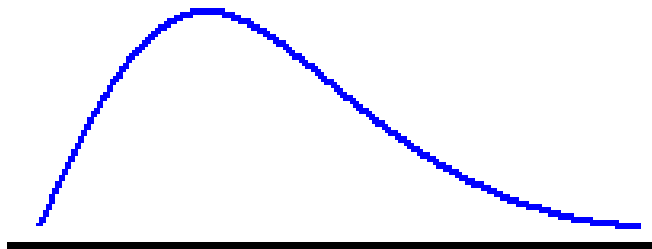
# Histogram Shapes



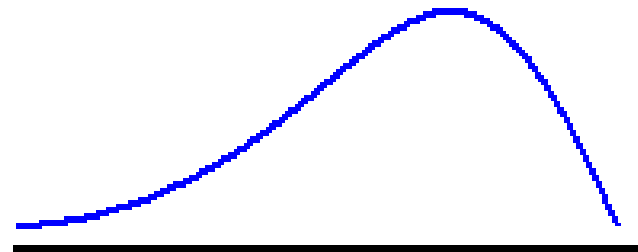
symmetric unimodal



bimodal



positively skewed



negatively skewed

# Measures of Location

# The Mean

The average (*mean*) of the  $n$  numbers  $x_1, x_2, \dots, x_n$  is  $\bar{x}$  where

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n}$$

*Population mean:  $\mu$*

# The Mean

The average (*mean*) of the  $\Sigma f_i$  numbers with frequencies

$$\bar{X} = \frac{f_1x_1 + f_2x_2 + \dots + f_nx_n}{f_1 + f_2 + \dots + f_n} = \frac{\sum f_ix_i}{\sum f_i}$$

# Median

The *sample median*,  $\tilde{x}$ , is the middle value in a set of data that is arranged in ascending order. For an even number of data points the median is the average of the middle two.

*Population median:  $\tilde{\mu}$*

# Median

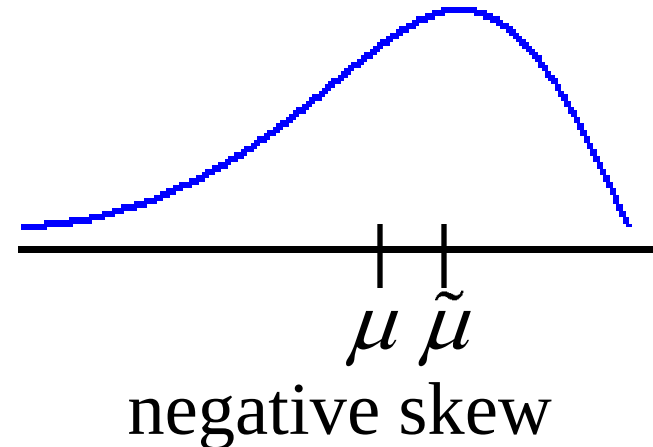
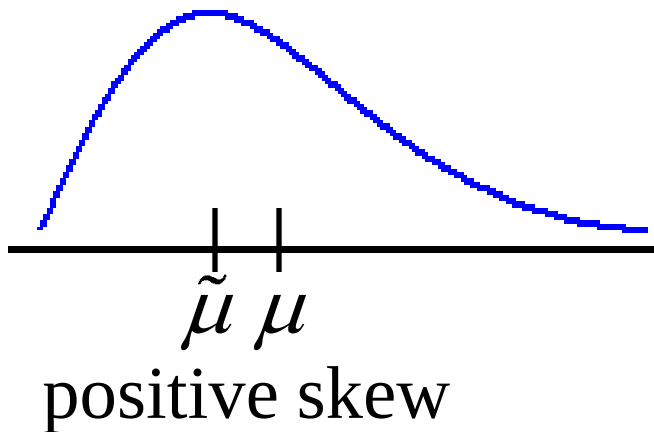
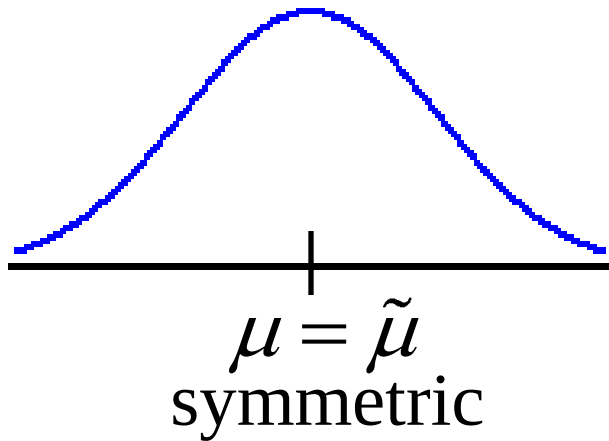
$$\tilde{x} = \begin{cases} x_{(n+1)/2} & , n \text{ odd} \\ \frac{1}{2} (x_{n/2} + x_{(n+1)/2}) & , n \text{ even} \end{cases}$$

# Mode

The *sample mode*,  $x_{\text{mode}}$  is the data which appears most frequently.

*Population mode*:  $\mu_{\text{mode}}$

# Three Different Shapes for a Population Distribution





# Measures of Variability

# Sample Variance

*Variance* is a measure of the spread of the data.

The *sample variance* of the sample  $x_1, x_2, \dots, x_n$  of  $n$  values of  $X$  is given by

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{S_{xx}}{n-1}$$

We refer to  $s^2$  as being based on  $n - 1$  degrees of freedom.

# Population Variance

*Variance* is a measure of the spread of the data.

The *population variance* of the population  $x_1, x_2, \dots, x_N$  of  $N$  values is given by

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

# Standard Deviation

*Standard deviation* is a measure of the spread of the data using the same units as the data.

The *sample standard deviation* is the square root of the sample variance, as well as *population standard deviation*:

$$s = \sqrt{s^2}$$

$$\sigma = \sqrt{\sigma^2}$$

## Formula for $s^2$

An alternative expression for the numerator of  $s^2$  is

$$\begin{aligned} S_{xx} &= \sum (x_i - \bar{x})^2 = \sum x_i^2 - \frac{(\sum x_i)^2}{n} \\ &= \sum x_i^2 - n\bar{x}^2 \end{aligned}$$

## Formula for $s^2$ and $\sigma^2$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$s^2 = \frac{n \sum x_i^2 - \left(\sum x_i\right)^2}{n(n-1)}$$

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

$$\sigma^2 = \frac{N \sum x_i^2 - \left(\sum x_i\right)^2}{N^2}$$

# Formula for $s^2$ and $\sigma^2$ for data with frequencies

$$s^2 = \frac{\sum f_i (x_i - \bar{x})^2}{n-1} \quad s^2 = \frac{n \sum f_i x_i^2 - \left( \sum f_i x_i \right)^2}{n(n-1)}$$
$$\sigma^2 = \frac{\sum f_i (x_i - \bar{x})^2}{N} \quad \sigma^2 = \frac{N \sum f_i x_i^2 - \left( \sum f_i x_i \right)^2}{N^2}$$

Ex: Determine the mean, median, mode, variance and standard deviation of given data.

| $x$ | $f$ |
|-----|-----|
| 50  | 6   |
| 75  | 8   |
| 80  | 12  |
| 90  | 10  |
| 100 | 4   |



Ex: Determine the mean, median, mode, variance and standard deviation of given data.

| $x$ | $f$ |
|-----|-----|
| 50  | 6   |
| 75  | 8   |
| 80  | 12  |
| 90  | 10  |
| 100 | 4   |

median:

$$\begin{aligned}\tilde{x} &= \frac{1}{2} (x_{n/2} + x_{(n+1)/2}) \\ &= \frac{1}{2} (x_{20} + x_{21}) = \frac{1}{2} (80 + 80) = 80\end{aligned}$$

mode:  $x_{\text{mode}} = 80$

Ex: Determine the mean, median, mode, variance and standard deviation of given data.

| $x_i$                        | $f_i$     | $f_i x_i$   | $(x_i - \bar{x})$ | $(x_i - \bar{x})^2$ | $f_i (x_i - \bar{x})^2$ | $x_i^2$ | $f_i x_i^2$   |
|------------------------------|-----------|-------------|-------------------|---------------------|-------------------------|---------|---------------|
| 50                           | 6         | 300         | -29               | 841                 | 5046                    | 2500    | 15000         |
| 75                           | 8         | 600         | -4                | 16                  | 128                     | 5625    | 45000         |
| 80                           | 12        | 960         | 1                 | 1                   | 12                      | 6400    | 76800         |
| 90                           | 10        | 900         | 11                | 121                 | 1210                    | 8100    | 81000         |
| 100                          | 4         | 400         | 21                | 441                 | 1764                    | 1000    | 40000         |
| <b><math>\Sigma =</math></b> | <b>40</b> | <b>3160</b> |                   |                     | <b>8160</b>             |         | <b>257800</b> |

Ex: Determine the mean, median, mode, variance and standard deviation of given data.

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{3160}{40} = 79$$

$$s^2 = \frac{\sum f_i (x_i - \bar{x})^2}{n-1} = \frac{8160}{40-1} = 209.2$$

$$s^2 = \frac{n \sum f_i x_i^2 - \left( \sum f_i x_i \right)^2}{n(n-1)} = \frac{40(257800) - 3160^2}{40(40-1)} = 209.2$$

$$s = \sqrt{s^2} = \sqrt{209.2} = 14.46$$

## Properties of Variance ( $s^2$ )

Let  $x_1, x_2, \dots, x_n$  be any sample and  $c$  be any nonzero constant.

1. If  $y_1 = x_1 + c, \dots, y_n = x_n + c$ , then  $s_y^2 = s_x^2$
2. If  $y_1 = cx_1, \dots, y_n = cx_n$ , then  $s_y^2 = c^2 s_x^2$ ,

where  $s_x^2$  is the sample variance of the  $x$ 's  
and  $s_y^2$  is the sample variance of the  $y$ 's.

# Upper and Lower Fourths (*Quartile*)

After the  $n$  observations in a data set are ordered from smallest to largest, the *lower* (*upper*) *fourth* (*quartile*) is the median of the smallest (largest) half of the data, where the median  $\tilde{x}$  is included\* in both halves if  $n$  is odd. A measure of the spread that is resistant to outliers is the *fourth spread*:

$$f_s = \text{upper fourth} - \text{lower fourth}.$$

\* Hoaglin, D.; Mosteller, F.; and Tukey, J. (Ed.). *Understanding Robust and Exploratory Data Analysis*. New York: Wiley, pp. 39, 54, 62, 223, 1983.

# Upper and Lower Fourths (*Quartile*)

Ex:

5, 6, 7, 8, 9, 10, 11, 12

$Q_1 = 6.5$ ,  $Q_2 = 8.5$ ,  $Q_3 = 10.5$

5, 6, 7, 8, 9, 10, 11, 12, 13

$Q_1 = 7$ ,  $Q_2 = 9$ ,  $Q_3 = 11$

5, 6, 7, 8, 9, 10, 11, 12, 13, 14

$Q_1 = 7$ ,  $Q_2 = 9.5$ ,  $Q_3 = 12$

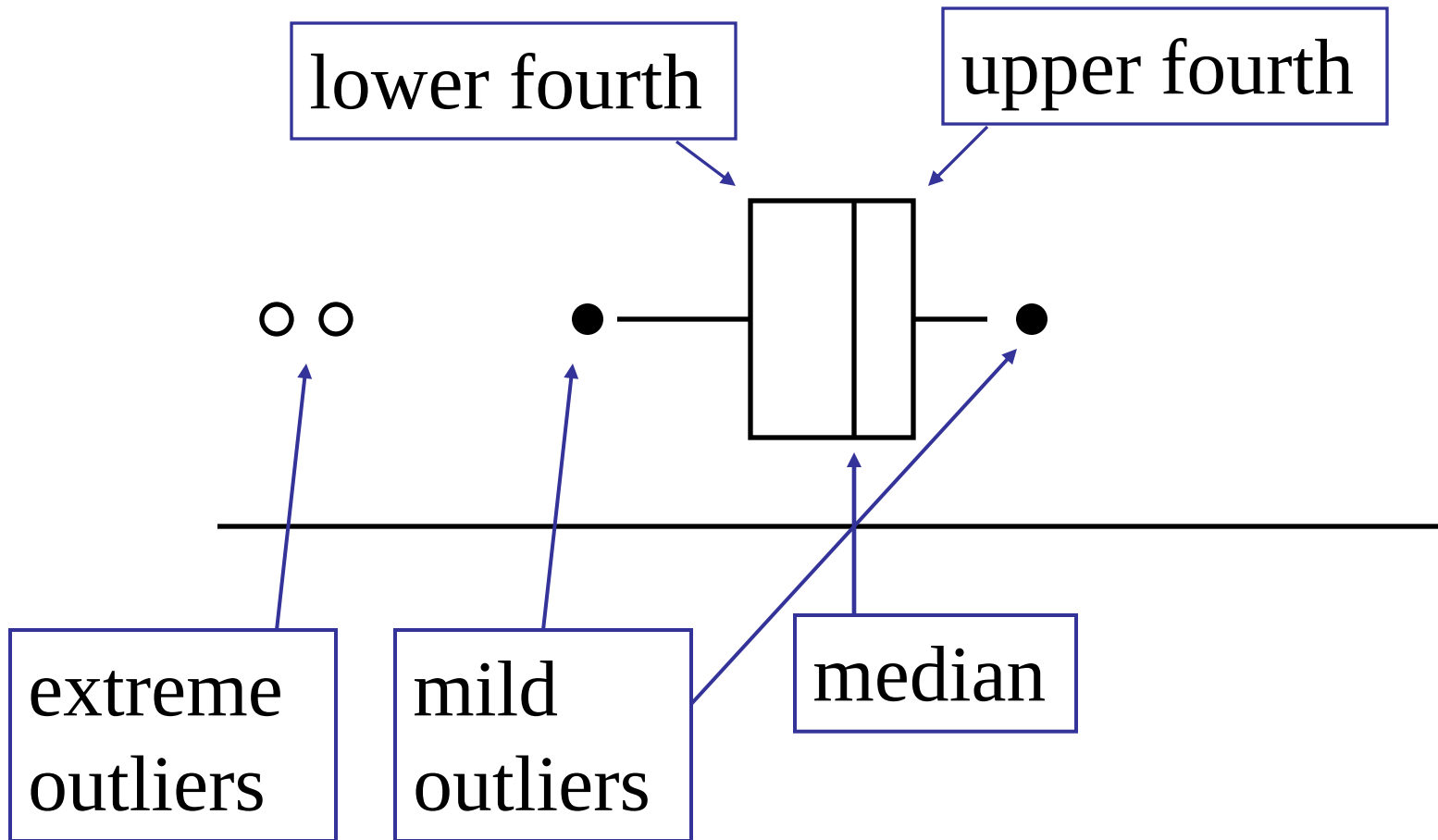
5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15

$Q_1 = 7.5$ ,  $Q_2 = 10$ ,  $Q_3 = 12.5$

# Outliers

Any observation farther than  $1.5f_s$  from the closest fourth is an *outlier*. An outlier is *extreme* if it is more than  $3f_s$  from the nearest fourth, and it is *mild* otherwise.

# Boxplots





# Formula for Grouped Data

Mean:

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} \quad x_i = \text{class midpoint}$$

Variance:

$$s^2 = \frac{\sum f_i (x_i - \bar{x})^2}{n-1} = \frac{n \sum f_i x_i^2 - \left( \sum f_i x_i \right)^2}{n(n-1)}$$

# Formula for Grouped Data

Median:

$$\tilde{x} = L + \frac{\frac{1}{2}n - f_k}{f} c$$

$x_i$  = class midpoint

$L$  = class lowerbound

$c$  = class interval length

$f$  = class frequency

$f_k$  = lower cumul.freq

$j = 1, 2, 3$

Fourth (*Quartiles*):

$$Q_j = L_j + \frac{\frac{j}{4}n - f_k}{f} c$$

$d_1$  = class freq – prev class freq

$d_2$  = class freq – after class freq

Mode:

$$x_{\text{mode}} = L + \frac{d_1}{d_1 + d_2} c$$

# Example of Grouped Data

| Data       | $f_i$     | midpoint $x_i$ | $f_i x_i$   | $x_i - \bar{x}$ | $(x_i - \bar{x})^2$ | $f_i(x_i - \bar{x})^2$ |
|------------|-----------|----------------|-------------|-----------------|---------------------|------------------------|
| 50 – 52    | 2         | 51             | 102         | –6.4            | 40.96               | 81.92                  |
| 53 – 55    | 4         | 54             | 216         | –3.4            | 11.56               | 46.24                  |
| 56 – 58    | 15        | 57             | 855         | –0.4            | 0.16                | 2.4                    |
| 59 – 61    | 6         | 60             | 360         | 2.6             | 6.76                | 40.56                  |
| 62 – 64    | 3         | 63             | 189         | 5.6             | 31.36               | 94.08                  |
| $\Sigma =$ | <b>30</b> |                | <b>1722</b> |                 |                     | <b>265.2</b>           |

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{1722}{30} = 57.4 \quad s^2 = \frac{\sum f_i (x_i - \bar{x})^2}{n-1} = \frac{265.2}{30-1} = 9.14 \quad s = \sqrt{s^2} = 3.02$$

# Example of Grouped Data

Mean: 
$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{1722}{30} = 57.4$$

Median: 
$$\tilde{x} = Q_2 = L + \frac{\frac{1}{2}n - f_k}{f} c = 55.5 + \frac{\frac{1}{2}30 - 6}{15} 3 = 57.13$$

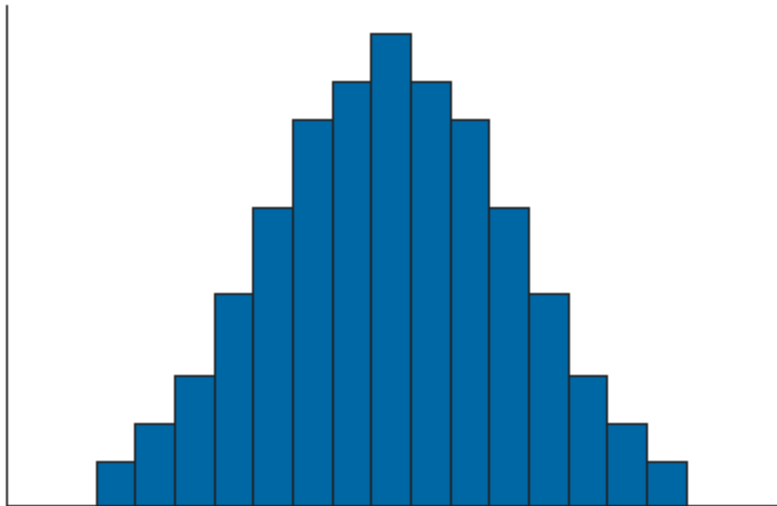
Mode: 
$$x_{\text{mode}} = L + \frac{d_1}{d_1 + d_2} c = 55.5 + \frac{(15 - 4)}{(15 - 4) + (15 - 6)} 3 = 57.15$$

1<sup>st</sup> quartile: 
$$Q_1 = L + \frac{\frac{1}{4}n - f_k}{f} c = 55.5 + \frac{\frac{1}{4}30 - 6}{15} 3 = 55.8$$

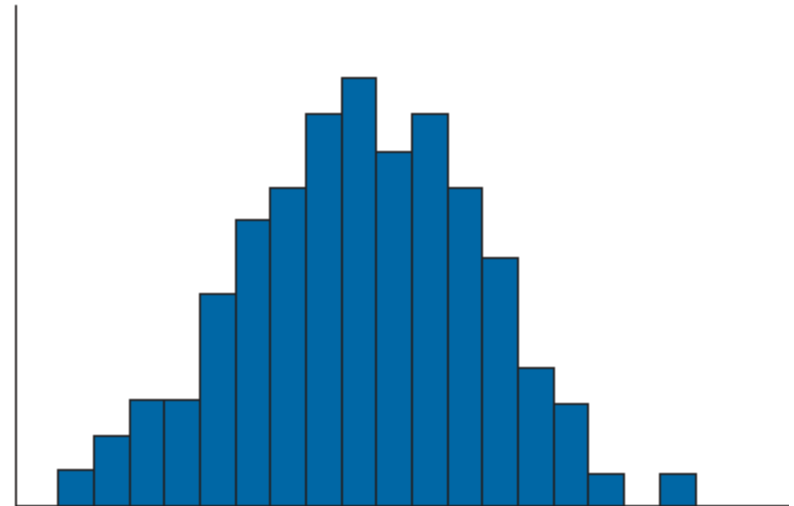
3<sup>rd</sup> quartile: 
$$Q_3 = L + \frac{\frac{3}{4}n - f_k}{f} c = 58.5 + \frac{\frac{3}{4}30 - 21}{6} 3 = 59.25$$

# Normal Data Sets

Normal:

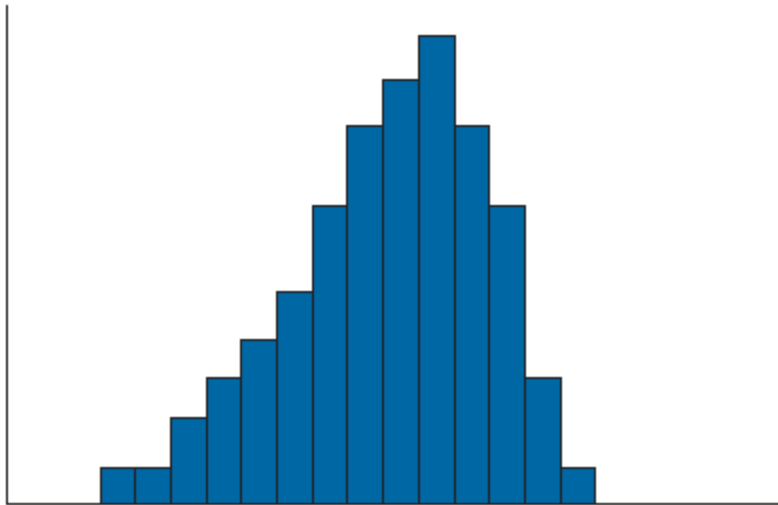


Approximately normal:

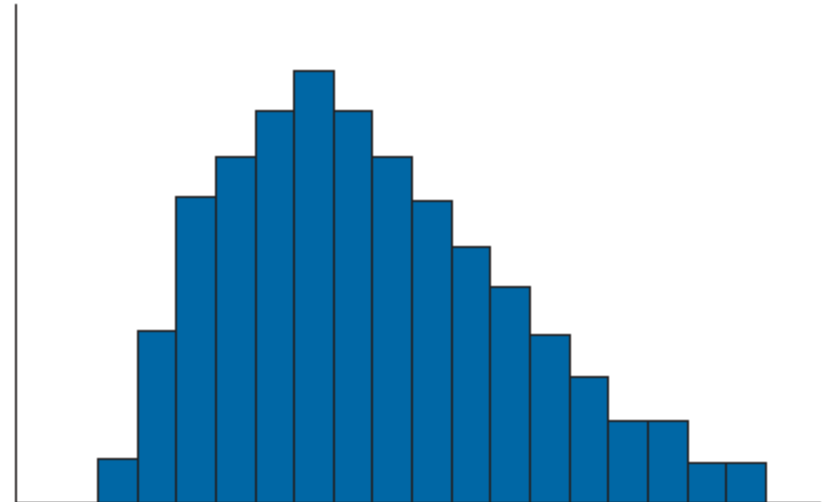


# Normal Data Sets

Left Skewed:



Right Skewed:



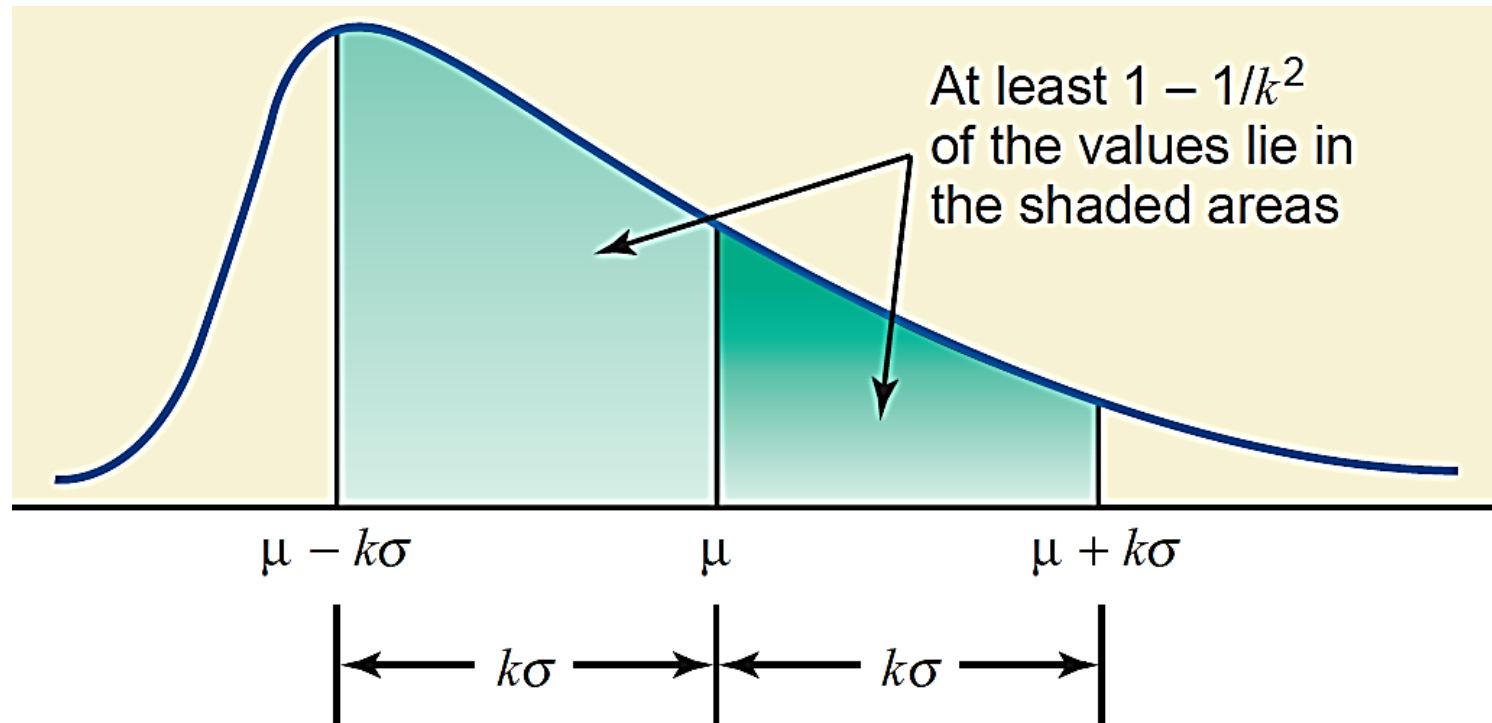
# Chebyshev's Inequality

The probability that any random variable  $X$  will assume a value within  $k$  standard deviations of the mean is at least  $1 - 1/k^2$ . That is,

$$P(|X - \mu| < k\sigma) > 1 - \frac{1}{k^2}$$

$$P(\mu - k\sigma < X < \mu + k\sigma) > 1 - \frac{1}{k^2}$$

# Chebyshev's Inequality

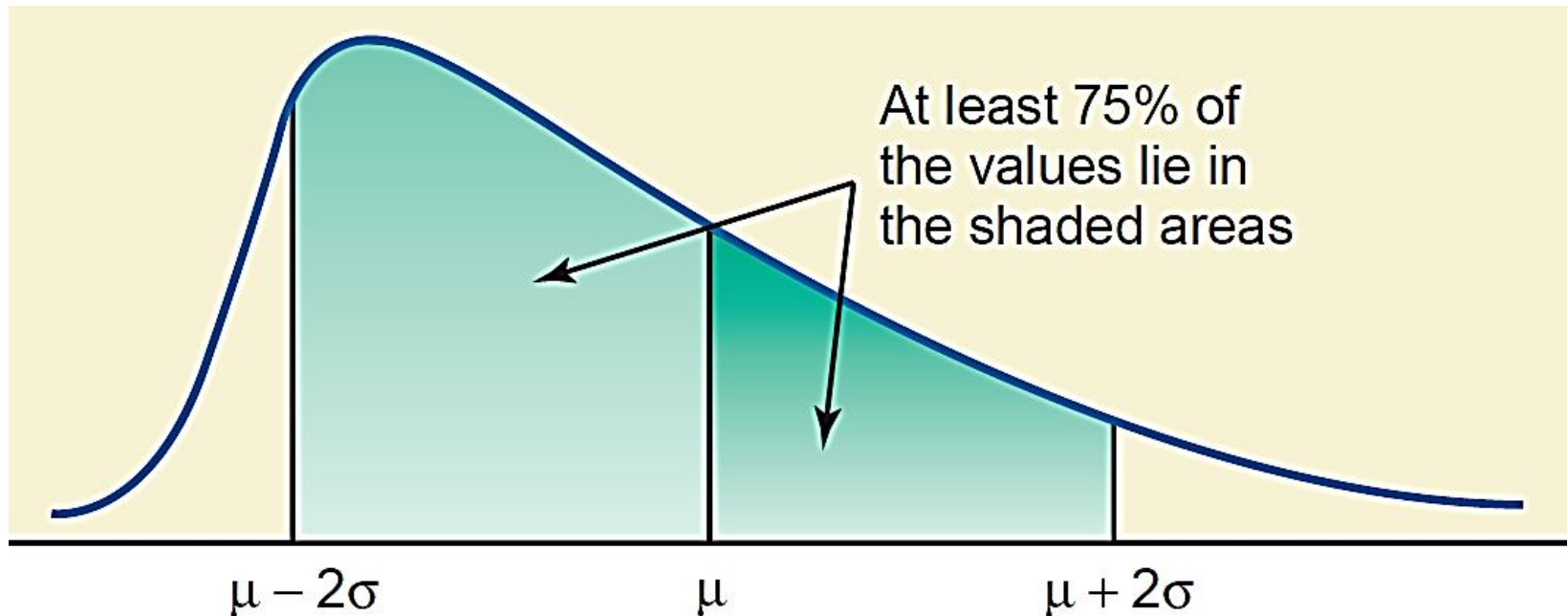


$$P(\mu - k\sigma < X < \mu + k\sigma) > 1 - \frac{1}{k^2}$$



# Chebyshev's Inequality

Ex.1. For  $k = 2$ , the theorem states that there is **at least**  $1 - 1/2^2 = 3/4 = 75\%$  of data points lie between  $\mu \pm 2\sigma$ .



# Chebyshev's Inequality

Ex. 2.

The average systolic blood pressure for 4000 women who were screened for high blood pressure was found to be 187 mmHg with a standard deviation of 22. Using Chebyshev's theorem, find at least how many women in this group have a systolic blood pressure between 143 and 231 mmHg.

# Chebyshev's Inequality

Ex. 2. Solution:

$$k\sigma = 187 - 143 = 231 - 187 = 44$$

$$k = 44/\sigma = 44/22 = 2$$

$$1 - 1/k^2 = 1 - 1/2^2 = 0.75$$

$$P(\mu - k\sigma < X < \mu + k\sigma) > 0.75$$

Therefore at least there are  $0.75 \times 4000 = \mathbf{3000}$  women in the group have a systolic blood pressure between 143 and 231 mmHg.



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